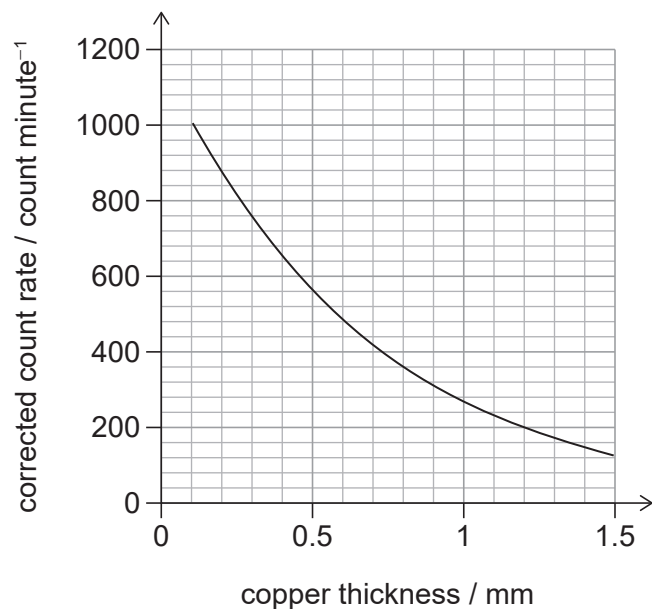


E.3 / PAPER 2 / QUESTION 2

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2. An experiment is carried out to determine the count rate, corrected for background radiation, when different thicknesses of copper are placed between a radioactive source and a detector. The graph shows the variation of corrected count rate with copper thickness.



- (a) Outline how the count rate was corrected for background radiation.

[1]

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- (b) When a single piece of thin copper foil is placed between the source and detector, the count rate is 810 count minute⁻¹. The foil is replaced with one that has three times the thickness. Estimate the new count rate.

[2]

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(Question 2 continued)

- (c) Further results were obtained in this experiment with copper and lead absorbers.

Absorber	Thickness / mm	Corrected count rate / count minute ⁻¹
copper	3.5	32
lead	3.5	10

Comment on the radiation detected from this radioactive source.

[4]

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- (d) Another radioactive source consists of a nuclide of caesium ($^{137}_{55}\text{Cs}$) that decays to barium ($^{137}_{56}\text{Ba}$).

Write down the reaction for this decay.

[2]

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